Three Jug Problem

Class 6: Weight and capacity

Introduction

The Three Jug Problem is a puzzle in which we are required to measure an exact quantity using jugs of fixed capacities. Since there are no markings on the jugs, we must rely on a sequence of well-defined actions: **fill, pour, and empty**.

Activity	Time	Description
Launch and Exploration	15 min	Ask the students “ If you have a 5L and a 3L jug, how can you measure exactly 4L? ” Explain the rules. For the activity, one can use Jugs proportional to the given capacities and can record actions like in the following: 1. Fill A 2. Pour A → B 3. Empty B Students try to solve (trial & error) Or can just represent the action on the grid provided (to mimic the water transfer) Template Isometric Grid Dots: https://tinyurl.com/threejugtemplate Demonstrate the solution
Discussions & Conclusion	20 min	Is this the only solution? Is there a solution that can be done shorter than the one provided in the demonstration ? Students work alongside the worksheet while the teacher guides the discussion. Student Worksheet: https://tinyurl.com/threejugworksheet Guide with prompts. Discuss the worksheet.

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CT Components

Algorithmic Thinking
Create a sequence of precise steps to reach the goal.

Decomposition
Break the problem into small actions (fill, pour, empty)

Pattern Recognition
Identify repeated cycles in their solution.

Abstraction
Representing the actions of fill, pour, and empty with the states in an isometric grid

Generalisation
Extending to other jug sizes

Evaluation
Compare different solutions based on the number of steps in the process, and on whether reaching a certain amount of water using jugs is possible.

Activity: Three Jug Problem



If we have a 5L and a 3L jug, how can we measure exactly 4L?

Rules:

- We may **fill**, **empty**, or **pour** between jugs
- No markings are allowed
- Jugs may be filled completely, emptied completely, or poured into another jug until one jug is full or the other is empty. Partial fills or partial pours are not allowed.

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Explain the rules to the students and let them try to solve by trial & error.

Demonstrate this to the student:

Step	Action	A (5L)	B (3L)
1	Fill the 3L jug from a water source (from Tap or Water-filled Buckets)	0	3
2	Transfer the water from 3L jug to 5L jug	3	0
3	Fill the 3L jug	3	3
4	Transfer the water from 3L jug to 5L jug	5	1
5	Empty the 5L jug	0	1
6	Transfer the water from 3L jug to 5L jug	1	0
7	Fill the 3L jug	1	3
8	Transfer the water from 3L jug to 5L jug	**4**	0

Once we follow this procedure step by step, we will get 4L of water into the 5L jug. Here, we have used 8 steps to find the solution.

Q. How many Fill, Transfer, and Empty steps were used in the above process?

a) 4, 3, 1
b) 1, 3, 4
c) 3, 4, 1
d) 3, 1, 4

Answer c) 3, 4, 1
In the 8 step solution (Fill 3L, Pour to 5L, Fill 3L, Pour to 5L, Empty 5L, Pour to 5L, Fill 3L, Pour to 5L), there are 3 Fills, 4 Pours (transfers), and 1 Empty.
CT Competency Algorithmic Thinking, Decomposition

Is this the only solution? Is there any solution that can be done shorter than the one provided in the demonstration?

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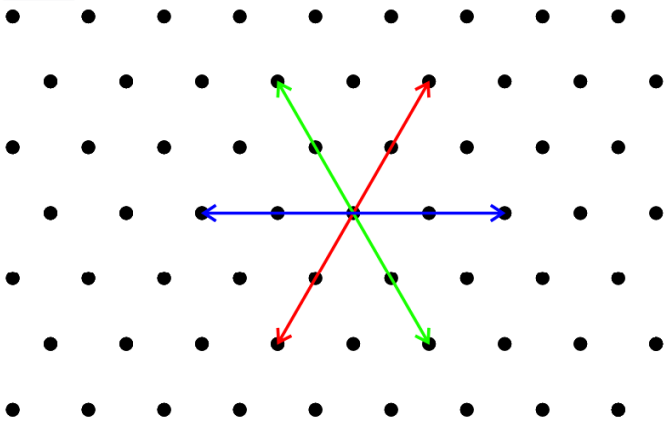
Let the students explore the alternative solution and guide them to observe whether it is shorter or longer than the previous one. Also, point out how they might revisit the same water amounts, creating loops that highlight the need for careful step tracking.

Another Solution :

Step	Action	A (5L)	B (3L)
1	Fill the 5L jug from a water source (from Tap or Water-filled Buckets)	5	0
2	Transfer the water from 5L jug to 3L jug	2	3
3	Empty the 3L jug	2	0
4	Transfer the water from 5L jug to 3L jug	0	2
5	Fill the 5L jug	5	2
6	Transfer the water from 5L jug to 3L jug	**4**	3

If we try to track the sequence of the various moves made above, it can be complicated. To keep track of all the moves above, we need a simpler way to represent them visually.

A useful way to do this is by using an isometric grid, which makes movement easier to see and record. Notice that in an isometric grid, there are three main directions of movement, as shown below. We can match these directions with our jug moves, such as moves related to Jug A and Jug B

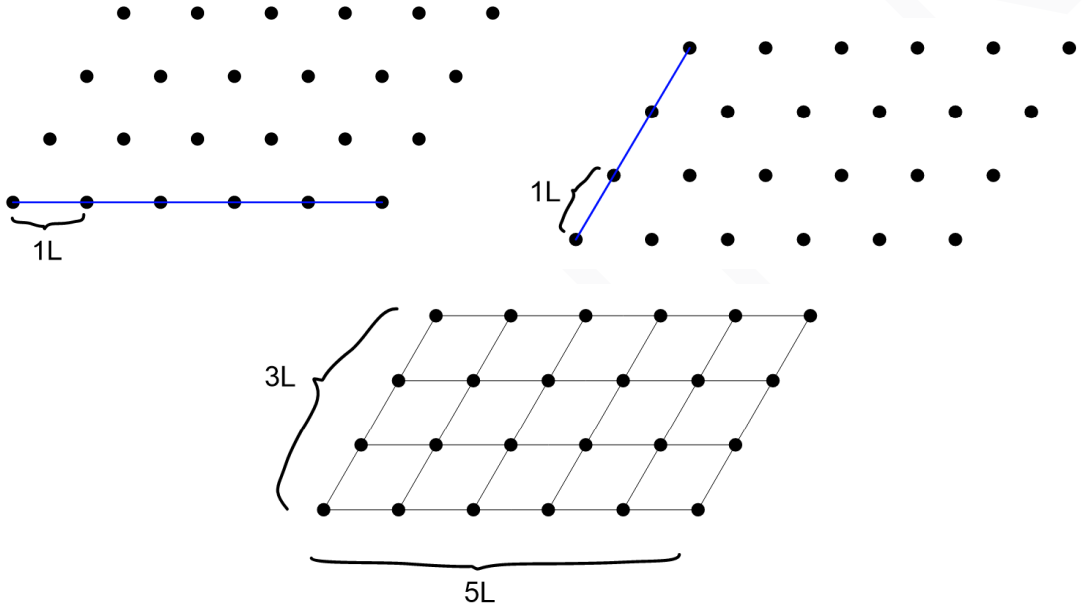


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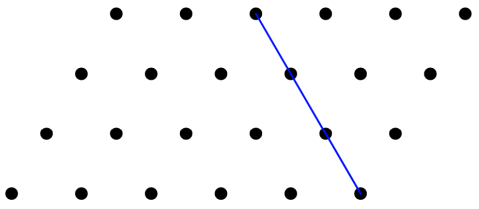
The horizontal line (Blue) represents only the pouring of water into Jug A. The right-slanted line (Red) represents the pouring of water into Jug B only. The left-Slanted Line (Green) represents the transfer of water between Jug A and Jug B.

Now we can use the isometric grid to represent the Fill, Transfer, and Empty actions. This will help us visualise the solution to the problem: If we have a 5L jug and a 3L jug, how can we measure exactly 4 litres?

Draw the 5 x 3 grid on the isometric grid template. We will represent a 5L jug by drawing five horizontal lines joining the dots. And three right-slanted vertical lines representing 3L jugs.

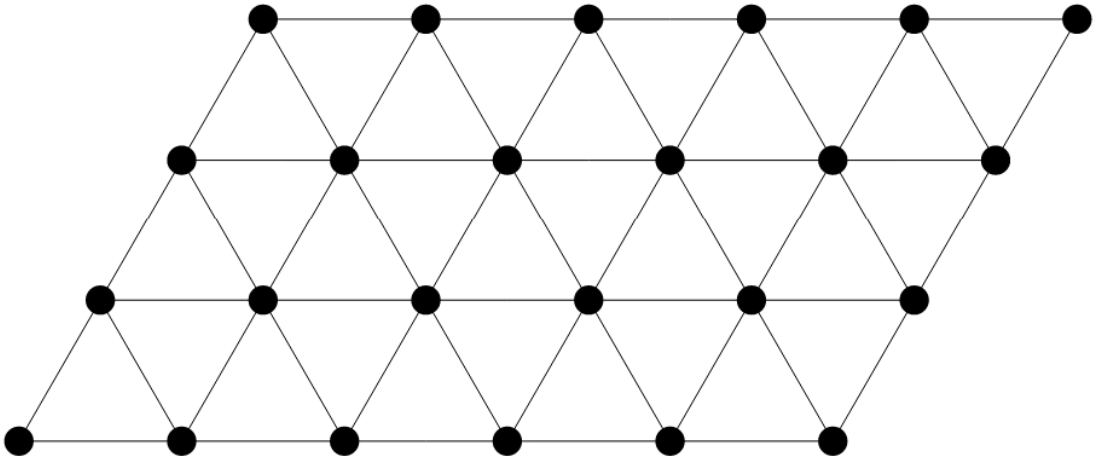


Also, we will join the dots with left-slanted vertical lines.



This will give us the final grid we need to work with.

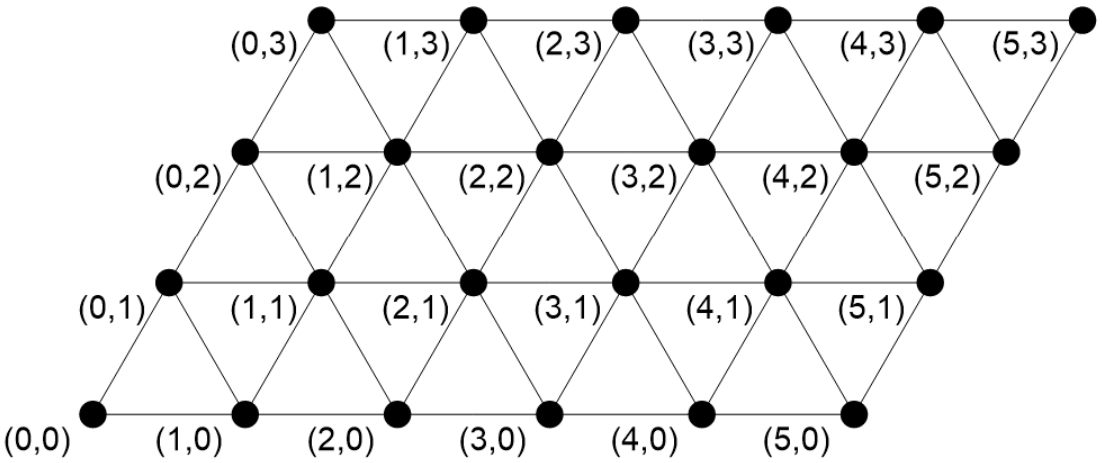
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The horizontal line represents only the pouring of water into the 5L Jug. The right-slanted line represents the pouring of water into the 3L Jug only. The left-Slanted Line represents the transfer of water between the 5L and 3L jugs.

We will represent each jug as a state (A, B) where A = amount of water in the 5L jug and B = amount of water in the 3L jug

(A, B) = (5L jug, 3L jug)

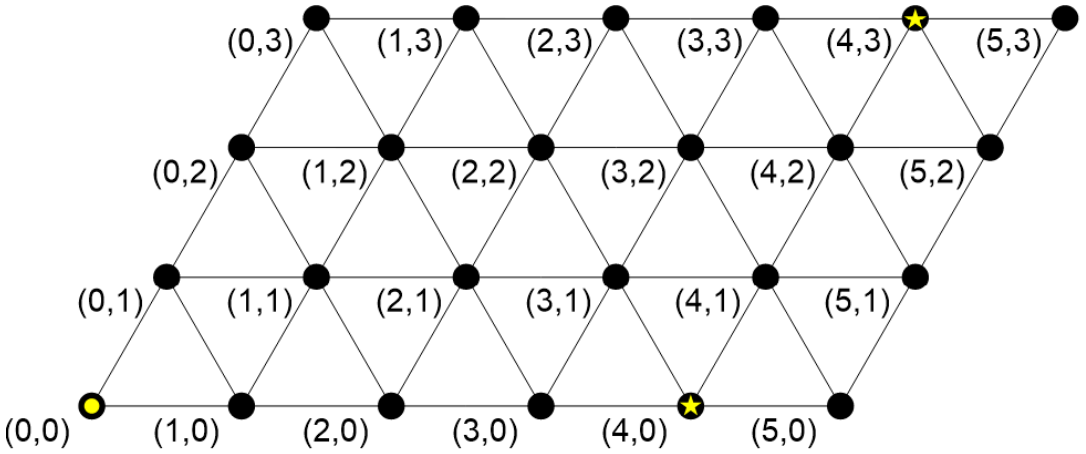


We have now translated the problem of filling, pouring, and emptying water into/from jugs into a path-finding problem on the grid.

Now, the objective is to find the path between Circle (Yellow) from any of the Stars (given) drawn in the grid, as they represent the 4L water point.

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Now, instead of pouring water, we simply need to draw the path from Start: (0, 0) to End: (4, 0) or (4, 3). And to mimic the complete pouring, transfer, and emptying into the jugs, we need to draw the line between the state till it hits the grid boundary.

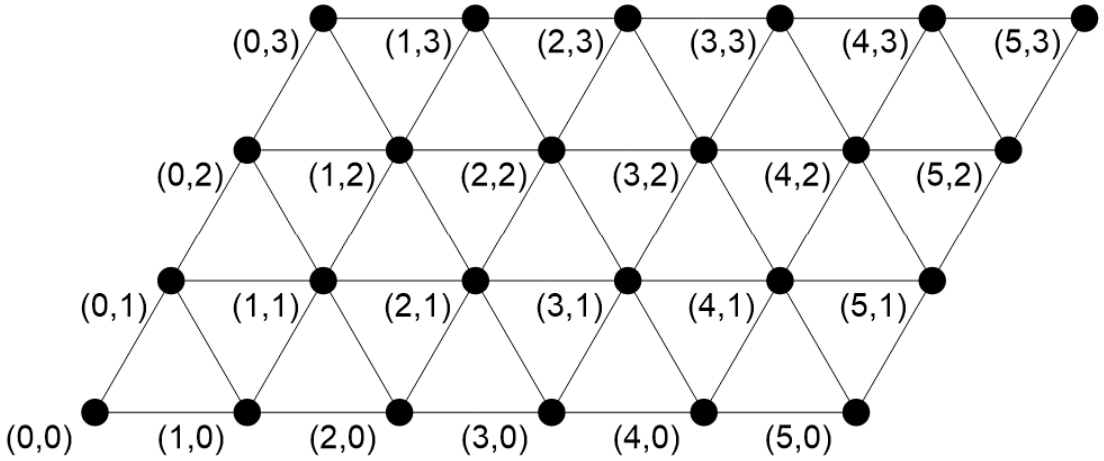


Let the student try to find the path.

Ask them to observe:
Why is reaching (4,1) and (4,2) not possible?

Answer
States (4,1) and (4,2) fall inside the grid, not on boundaries. While they contain the target 4L total, the rules prevent reaching them.
CT Competency Abstraction, Evaluation

After student trials, introduce grid states gradually: have them solve practice questions first to build familiarity, before demonstrating the full solution on the grid.



Q. If you reach the state (5, 1), meaning the 5L jug is full, and the 3L jug has 1L, and your next action is "Empty the 5L jug," what is your new state (A, B)?
a) (5, 0)
b) (0, 1)

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c) (0, 0)
d) (3, 3)

Answer b) (0, 1)
Empty 5L: New state is 0,1.
CT Competency Abstraction, Logic, Decomposition

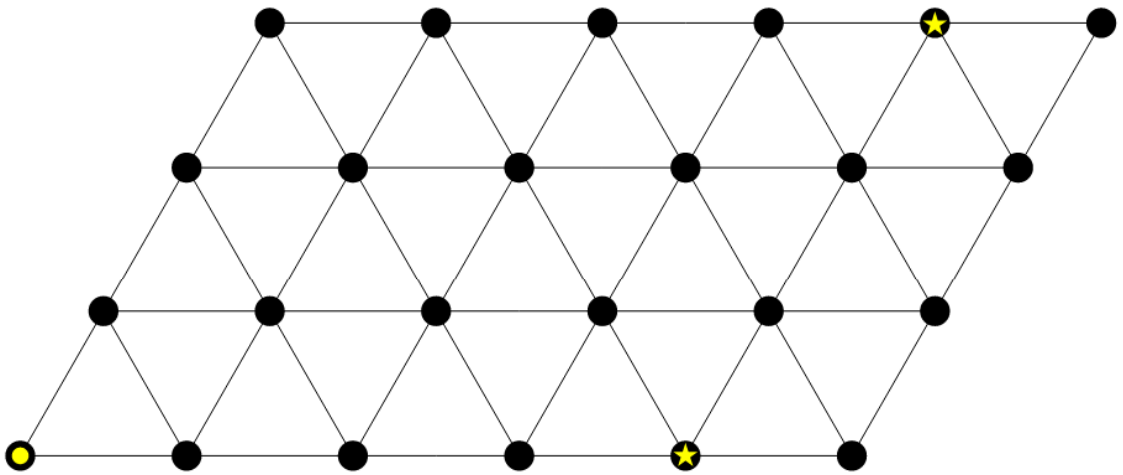
Q. You are at state (2, 3), meaning the 5L jug has 2L, and the 3L jug is full. If your next action is "Empty the 3L jug," what is the new state (A, B)?
a) (0, 3)
b) (5, 2)
c) (2, 0)
d) (0, 0)

Answer c) (2, 0)
Empty 3L: New state is 2,0.
CT Competency Abstraction, Logic, Decomposition

Q. Moving from state (0, 3) to state (3, 0) represents which action?
a) Transfer water from the 5L jug to the 3L jug.
b) Empty the 3L jug.
c) Transfer water from the 3L jug to the 5L jug.
d) Fill the 3L jug from the source.

Answer c) Transfer water from the 3L jug to the 5L jug.
0,3 → 3,0 Action: This is a transfer (pour) from 3L to 5L.
CT Competency Decomposition, Abstraction, Logic

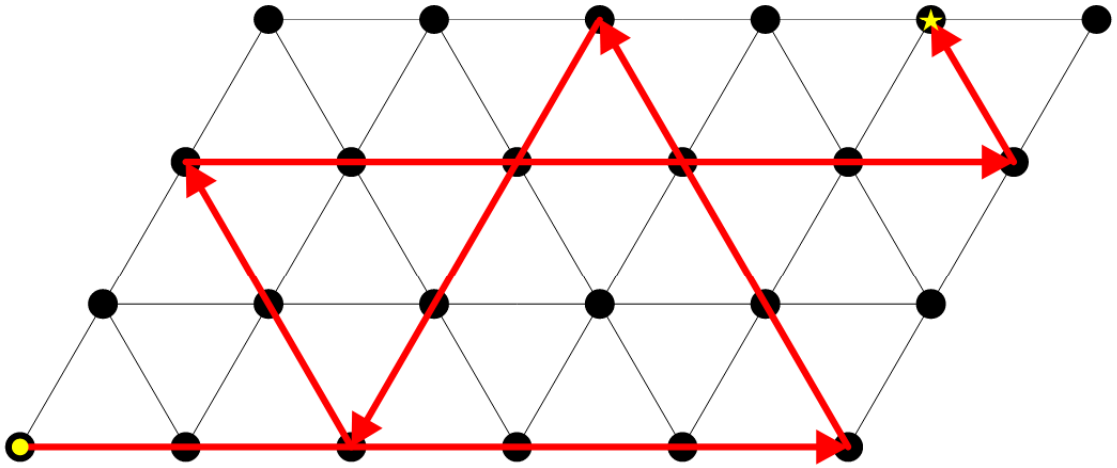
Draw a path in the grid below to measure exactly 4L using 5L and 3L jugs?



Did the path take the same steps as you wrote earlier in the table format?

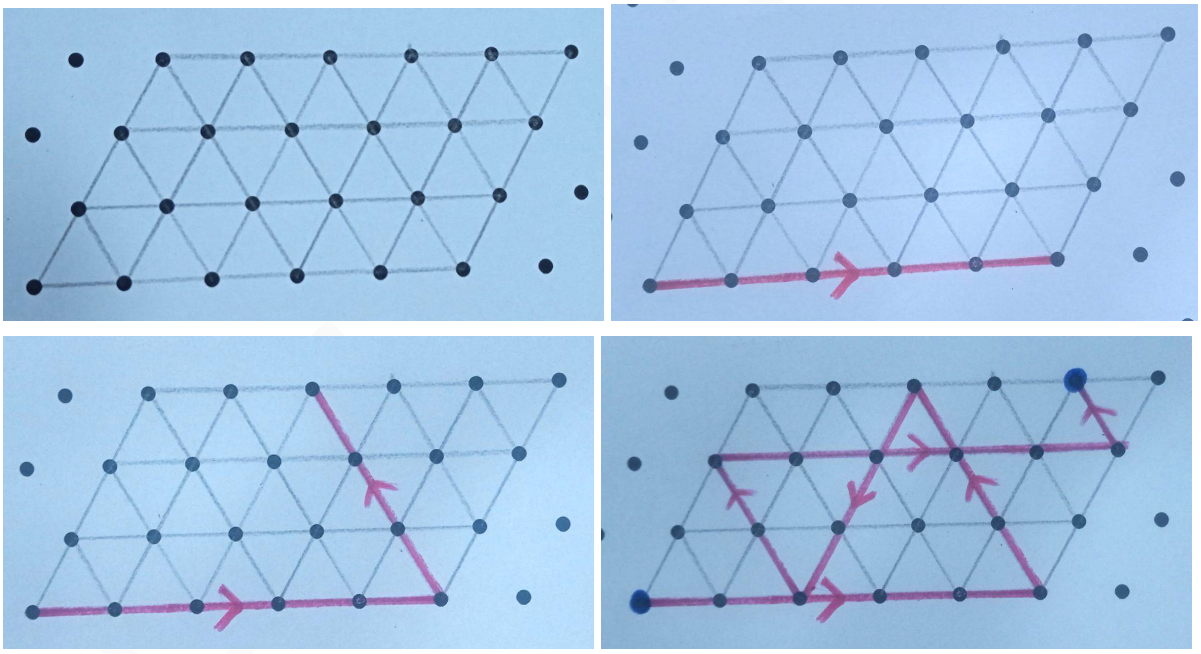
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Answer



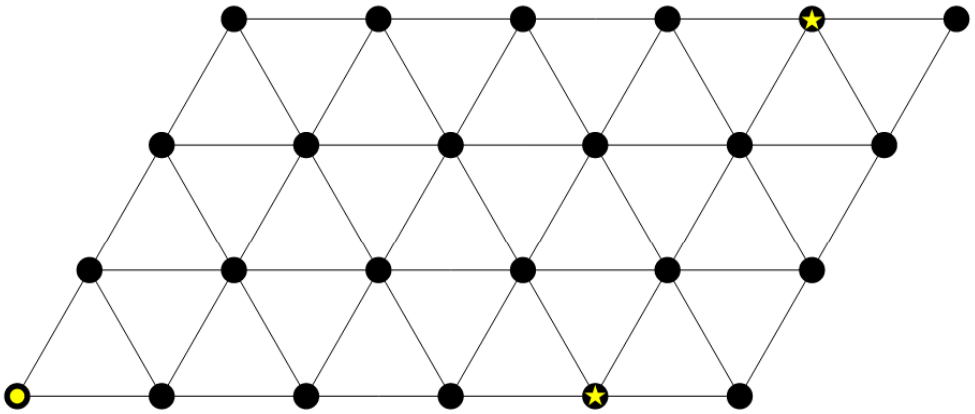
Yes, the steps for filling, pouring, and emptying the jugs are the same as the steps on the path taken.

CT Competency Algorithmic Thinking, Decomposition, Abstraction

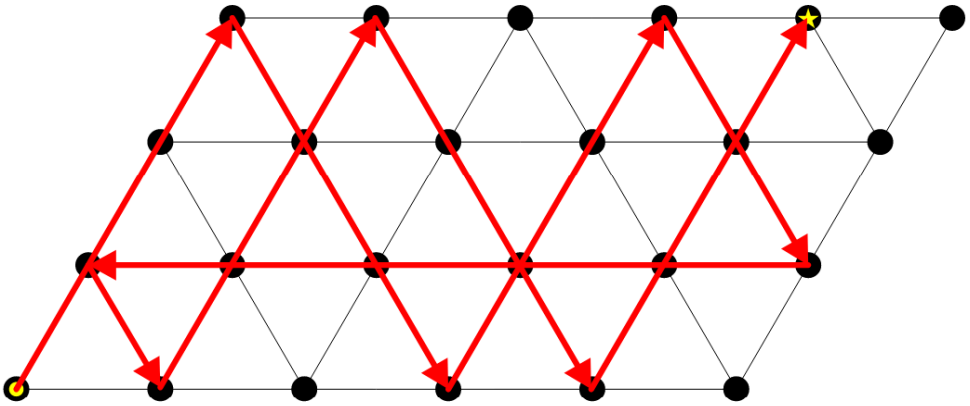


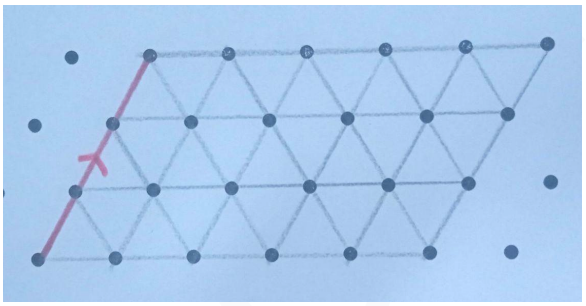
Draw another solution.

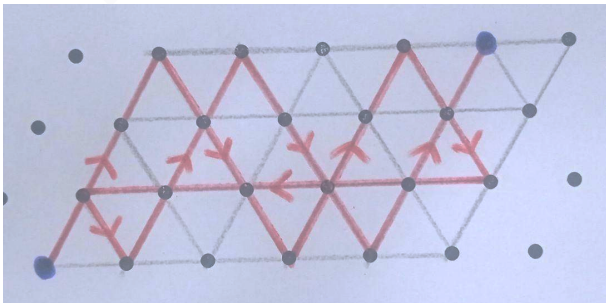
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Another Solution

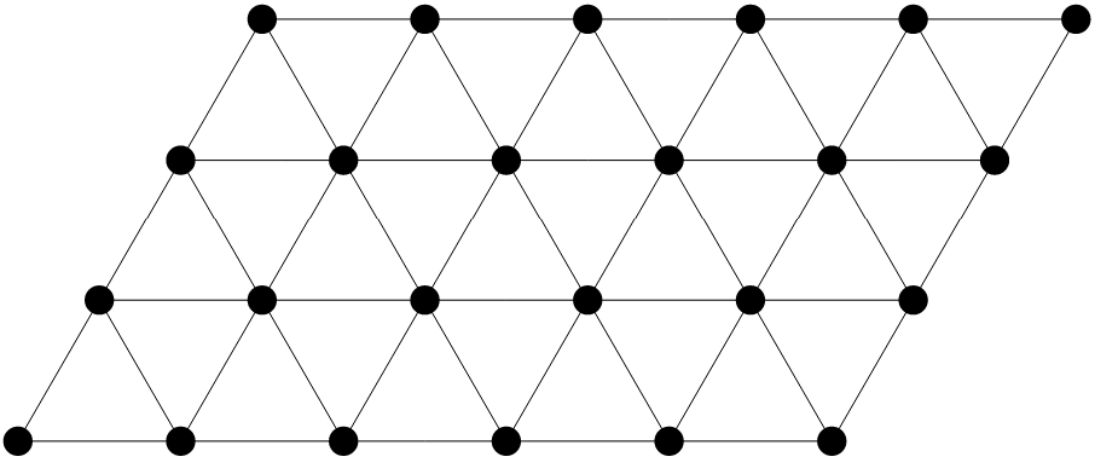




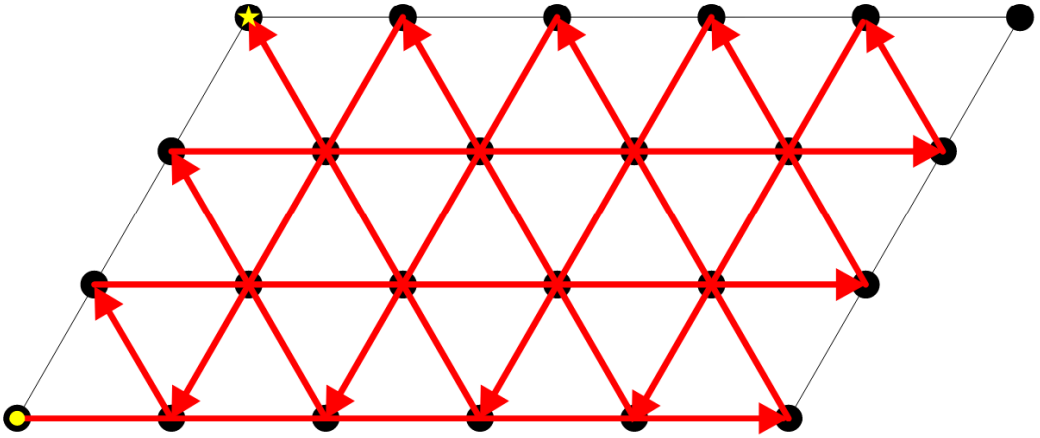


Q. Draw a path in the grid below that visits every dot.

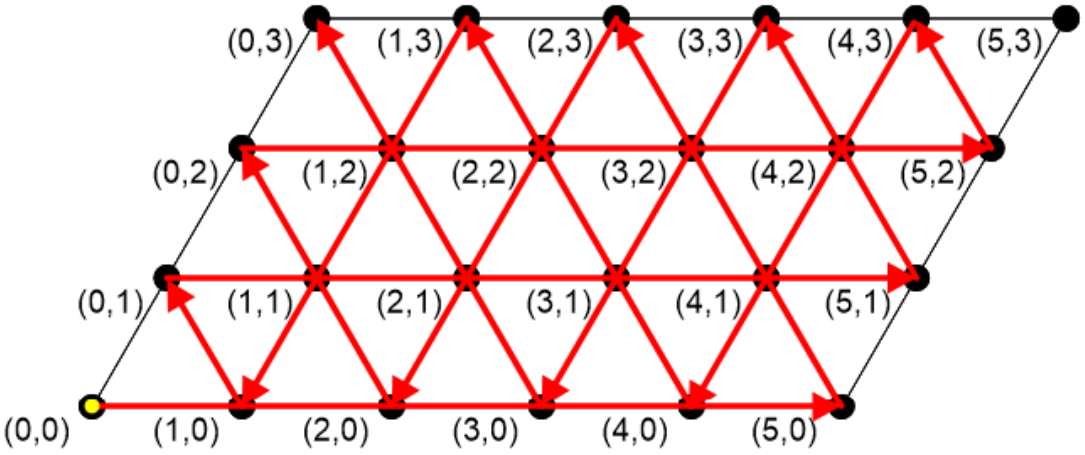
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Solution



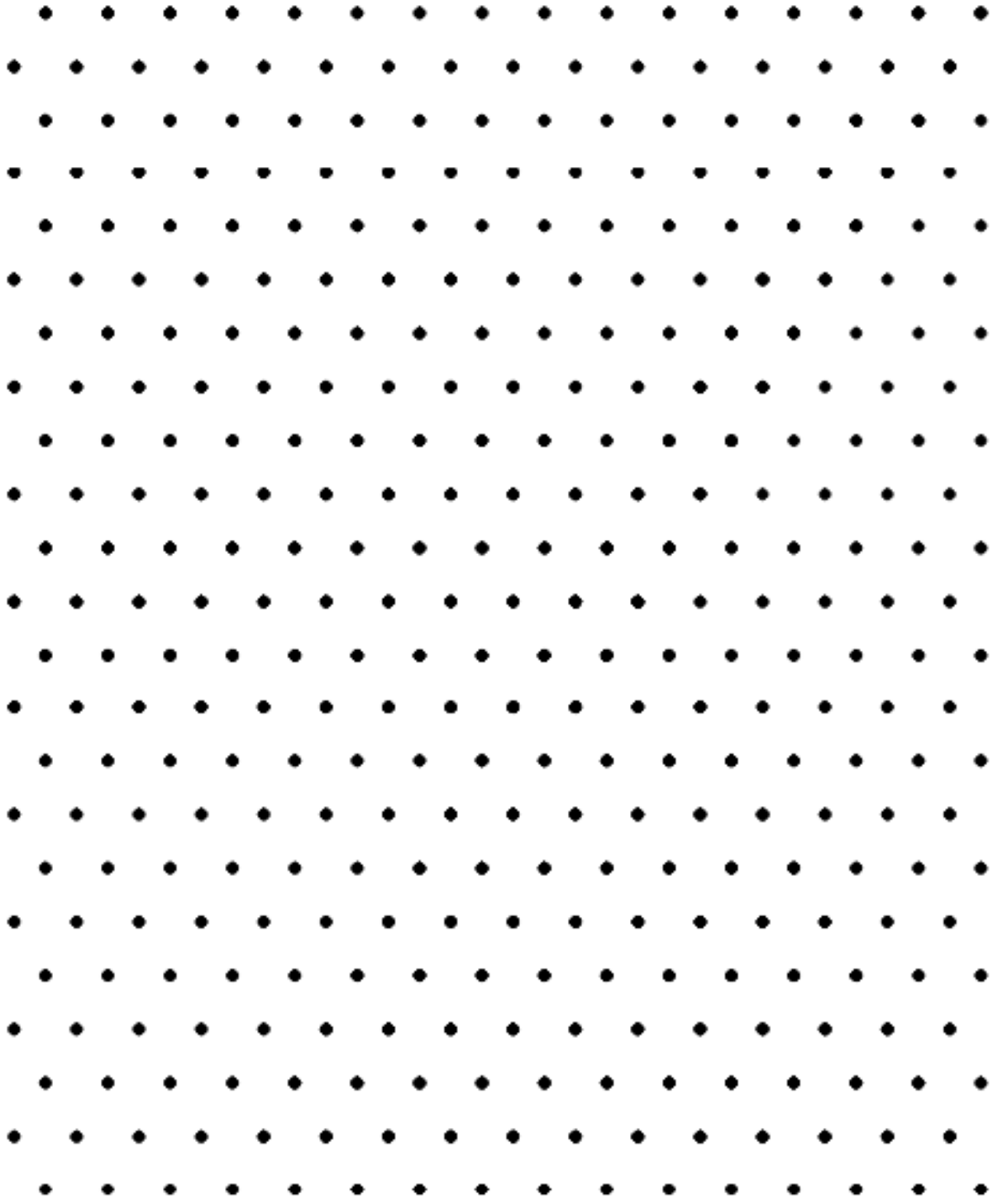
With this, we can observe that with 5L and a 3L jug, we can obtain exact quantities of 1L, 2L, 3L, 4L, and 5L, and if we consider adding quantities from both jugs, we can obtain 6L, 7L, and 8L jugs.



Now, with the same rules, if you have a 4L and a 3L jug, how can you measure exactly 2L?

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Also, measure 6L using an 8L and a 5L jug.



In this activity, we have solved the Three Jug Problem by creating algorithms from simple actions (fill, pour, empty), decomposing complex measurements into states, abstracting to numbers/grids, and evaluating efficient paths.

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